

SEV-Cap

Semantic-Entropy Verified Video Captioning

Hackathon Track 2 · Video Captioning
Four styles · Verified facts · Anytime output

Live Demo

associates-desired-avatar-doctrine.trycloudflare.com

github.com/skx56/sev-cap
Docker: ghcr.io/skx56/sev-cap:latest

The Problem

Single-pass video captioners — one VLM call, four styles — fail the LLM-Judge on two predictable axes:

- **Accuracy:** models invent details not visible in the clip (hallucinated objects, actions, dialogue).
- **Tone:** a caption labeled sarcastic often reads as plain description — the style label is aspirational, not earned.

Why it matters

- The judge scores each caption on accuracy (1–5) and tone (1–5).
- Pipelines that cannot detect their own hallucinations leave points on the table.
- Hackathon containers time out — missing output scores zero.

SEV-Cap attacks both failure modes structurally — with verification gates the judge actually cares about, not a bigger prompt.

Four Required Outputs

Every clip produces exactly four captions — one per mandated style:

Formal	Clear, professional scene description
Sarcastic	Dry, witty take with a bit of attitude
Humorous · Tech	Playful joke with a nerdy / tech twist
Humorous · Non-Tech	Light, everyday humor anyone can follow

Two-Phase Pipeline

- **Phase 1 — Draft:** fast vision pass writes all four styles immediately. Every clip has valid output from second one.
- **Phase 2 — SEV Upgrade:** sample facts, verify with semantic entropy, regenerate with gates. Keep draft unless both gates pass.

Stage 1 — Verified Facts

Evidence Collection

- **Keyframes:** ffmpeg samples uniform + scene-change frames.
- **Audio:** faster-whisper transcribes locally (CPU, baked into Docker).
- Dialogue and sound cues become facts that vision alone would miss.

Semantic-Entropy Verification

- K=3 independent vision extractions (frames + transcript).
- Atomic facts clustered by bidirectional entailment (LLM as NLI judge).
- Per-fact support across samples gives a semantic-entropy signal.
- Facts in 2 of 3 samples are **verified**; singletons are rejected.
- Based on Farquhar et al., Nature 2024 — confabulation signature.

Rejected high-entropy facts are logged in every clip's verification report.

Stage 2 — Grounded Generation

Four style generators rewrite captions from the verified fact sheet **and** keyframes. Vision context prevents fact-sheet-only captions from missing story beats.

Gate A — Grounding

- Every concrete claim must be entailed by the verified fact sheet.
- Failures trigger Self-Refine with offending claims as feedback.

Gate B — Blind Style Lineup

- All four captions are label-stripped, shuffled, and re-identified by a fresh judge (confidence must be 3/5 or higher).
- A sarcastic caption that reads declarative loses and is rewritten.
- One Self-Refine round (Madaan et al., NeurIPS 2023).

Draft firewall: upgraded captions replace the draft only when they pass *both* gates. Otherwise the reliable vision draft is kept.

Built for the Scoring Harness

Anytime Algorithm

- Phase 1 writes valid JSON for every clip before Phase 2 starts.
- Upgrades overwrite atomically — timeouts degrade quality, never produce missing output.
- Per-clip upgrade timeout (300s) + global time budget (1800s).

Docker Container

- linux/amd64 image on ghcr.io — argument-free entrypoint.
- Whisper weights baked at build time (no extra network at runtime).
- FIREWORKS_API_KEY injected by harness; never baked into image.
- CI: pytest on push, then build and publish via GitHub Actions.

Model Strategy

- **Default:** Kimi K2.6 (Fireworks serverless) — extractor, judge, writer, refiner.
- **Bonus mode:** Gemma 4 26B on-demand via env override, with Kimi fallback on cold start or degenerate output.

Architecture

Video clip

```
|-- ffmpeg -----> keyframes
|-- faster-whisper ----> transcript
|
|-- Phase 1: vision draft (4 styles) --> atomic write to disk
|
+-- Phase 2: K=3 independent extractions
    |-- bidirectional entailment clustering
    |-- semantic entropy (keep facts with 2/3 support)
    |-- verified fact sheet
    |-- 4 style generators (+ keyframes)
    |   |-- Gate A: grounding check
    |   |-- Gate B: blind style lineup
    |   +-- Self-Refine (1 round)
    |       |-- both pass --> upgraded output
    +-----+-- fail/timeout --> keep draft
```

Every clip JSON includes a verification report: verified facts, rejected high-entropy facts, lineup verdicts, and retry history.

Results and Tech Stack

Internal Eval (7 clips, LLM-Judge)

Metric	Score
Mean accuracy (1–5)	3.82
Mean tone (1–5)	4.79
Combined (leaderboard 0–1)	0.86

Tech Stack

- Kimi K2.6 (default) · Gemma 4 26B (bonus) · Fireworks AI API
- faster-whisper · ffmpeg · Python 3.11 · asyncio
- Streamlit demo · Docker linux/amd64 · GitHub Actions to ghcr.io

Try It Live

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Upload a clip, get four verified captions, inspect the fact report.